

Paper 3 Answers

- 1 (a) (i) MgO and Al₂O₃ have giant ionic structures consisting of cations and anions held together by strong ionic bonds, requiring large amount of energy to break the strong bonds.

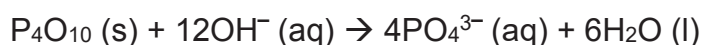
Al³⁺ in Al₂O₃ has a smaller ionic radius than Mg²⁺ but has a higher ionic charge of 3+ thus have a high charge density and high polarising power.

It is able to cause some polarisation of the O²⁻ ion electron cloud, giving rise to some covalent character / weakening the ionic bond.

SO₃ has a simple molecular structure consisting of SO₃ molecules held together by weak dispersion forces thus require small amount of energy to break the bonds.

- (ii) Large electronegativity difference in Mg and O cause the bonds in MgO to be ionic and a basic oxide is obtained.

Smaller electronegativity difference in P and O causes the bond in P₄O₁₀ to be covalent hence acidic oxide is obtained.



- (ii) In Sample 1: 7 moles of **C** (Na₂O) (in Sample 1) dissolved in water to produce 14 mol of OH⁻ ions which reacted with 12 mol of H⁺ from dissolution of 1 mol of P₄O₁₀ and 1 mol of Al₂O₃ to give a neutral solution.

In Sample 2: 1 mol of **A** (P₄O₁₀) reacts with water to produce 12 mol of H⁺, 6 mol of which reacted with 1 mol of Al₂O₃ and 2 mol with 2 mol NaOH formed, with 4 mol of H⁺ still remaining in solution.

A & B: Al₂O₃ & P₄O₁₀ **C:** Na₂O



- (ii) Mg²⁺ has a smaller ionic radius than Ca²⁺ while their ionic charges are the same. Charge density of Mg²⁺ is higher and hence has higher polarising power than Ca²⁺. Mg²⁺ is able to polarise / distort the electron cloud of CO₃²⁻ causing MgCO₃ to be less thermally stable and decompose at 315 °C. Both carbonates will decompose at a higher temperature of 530 °C hence volume of CO₂ evolved is doubled.

(iii)

$$n_{\text{pure dolomite present}} = \frac{0.450 / 44.0}{2} = 5.11 \times 10^{-3} \text{ mol} \quad [1]$$

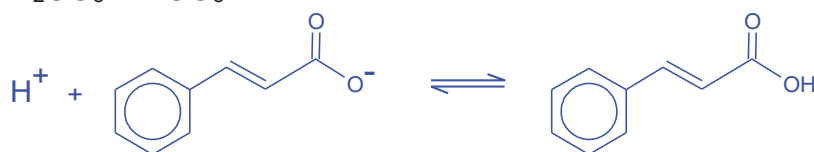
$$\% \text{ purity of dolomite} = \frac{5.11 \times 10^{-3} \times (40.0 + 24.0 + 2(12.0 + 3 \times 16.0))}{1.000} \times 100$$

$$= 94.09 \approx 94.1\% \quad [1]$$

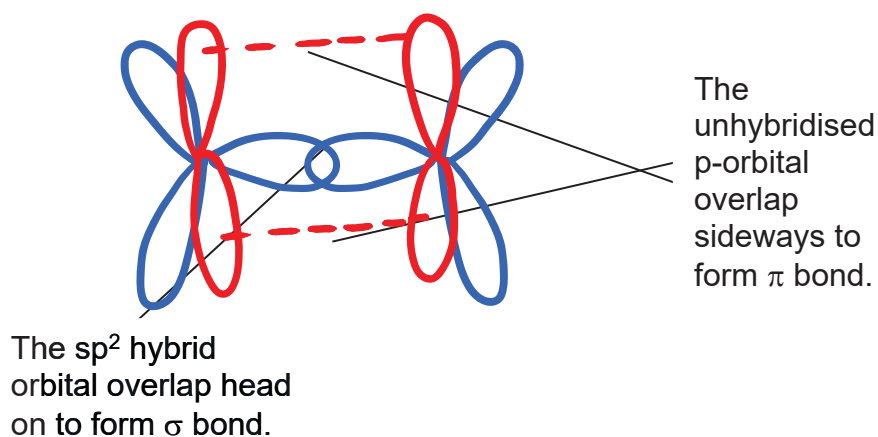
- (d) (i) Cinnamic acid will react with aq NaOH first. The p orbital of O atom in cinnamate ion overlaps with the pi orbital of the C=O , C=C and benzene to form a delocalised electron cloud. The negative charge on O of cinnamate ion is dispersed into the delocalised cloud, stabilising the ion hence is more acidic.

The ethanoate ion has an electron donating methyl group that increases the electron density of the CH_3CO_2^- ion hence destabilising it.

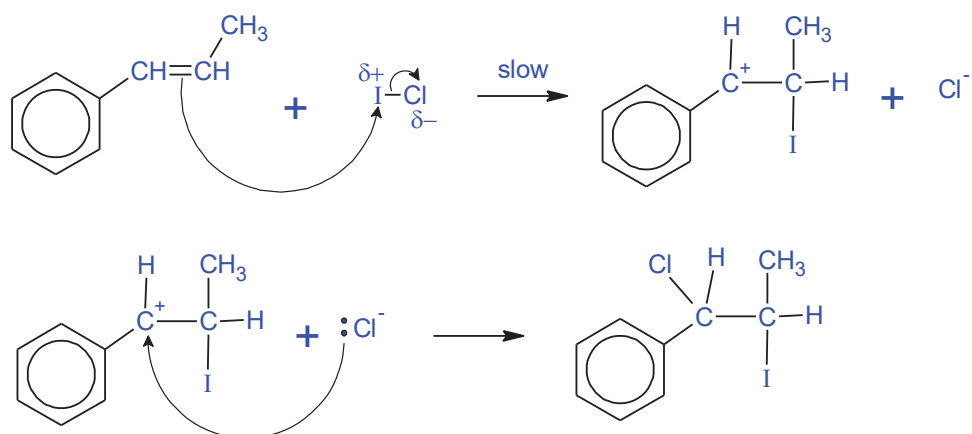
- (ii) CO_2 will dissolve in water to form carbonic acid.
 $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$
 The carbonic acid is a stronger acid than cinnamic acid thus will dissociate in water to and protonate the cinnamate ion to form back cinnamic acid.
 $\text{H}_2\text{CO}_3 \rightleftharpoons \text{HCO}_3^- + \text{H}^+$



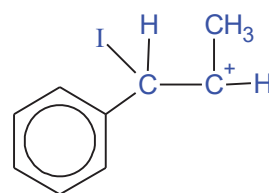
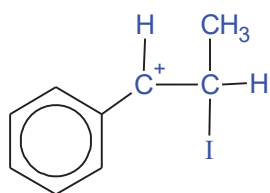
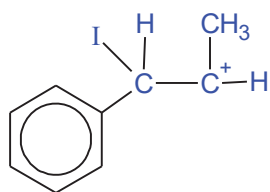
2 (a)



- (b) (i) Electrophilic Addition

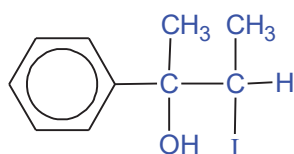


(ii)



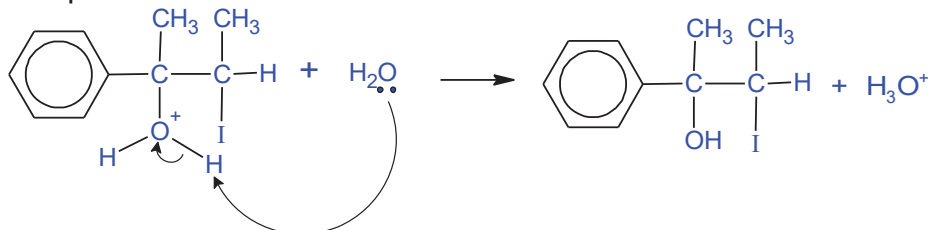
is more stable than and hence is the preferred carbocation formed. The p orbital of the carbocation centre overlaps with the delocalised π electron cloud of the benzene ring, electrons from the benzene ring delocalises to the carbocation centre. Hence the positive charge can be dispersed into the benzene ring, stabilizing it further.

(iii)



The lone pairs of electrons on O in H₂O acts as competing **nucleophile** in the 2nd step of the mechanism to attack the carbocation intermediate.

Note: A proton will be lost subsequently to form the organic compound as shown in the structure above.



(c)

	$\text{C}_6\text{H}_5\text{CH}_2\text{CH}=\text{CH}_2$	CO	H_2	$\rightleftharpoons$	$\text{C}_6\text{H}_5(\text{CH}_2)_3\text{CHO}$
Initial pressure / atm	40	40	40		0
Change in pressure / atm	-39.6	-39.6	-39.6		+39.6
Equilibrium pressure / atm	0.4	0.4	0.4		39.6

$$K_p = \frac{(P_{\text{C}_6\text{H}_5\text{CH}_2\text{CH}=\text{CH}_2})}{(P_{\text{C}_6\text{H}_5(\text{CH}_2)_3\text{CHO}})(P_{\text{CO}})(P_{\text{H}_2})} = \frac{39.6}{(0.4)(0.4)(0.4)} = 619 \text{ atm}^{-2}$$



amount of H_2 , n_{H_2}

$$= \frac{10.9 \times 10^{-3}}{22.7} = 4.80 \times 10^{-4} \text{ mol}$$

amount of $C_xH_yOH = 2 \times n_{H_2}$

$$= 2 \times 4.80 \times 10^{-4} = 9.60 \times 10^{-4} \text{ mol}$$

Contraction = initial total volume – final total volume

$$65 = (V_{O_2 \text{ total}}) - (V_{O_2 \text{ excess}} + V_{CO_2})$$

$$V_{O_2 \text{ total}} - V_{O_2 \text{ excess}} = 65 + V_{CO_2}$$

$$V_{O_2 \text{ reacted}} = 65 + V_{CO_2} = 65 + 131 = 196 \text{ cm}^3$$

OR

Let initial V_{O_2} be $z \text{ cm}^3$.

	C_xH_yOH (l)	$(\frac{4x+y-1}{4})$ O_2 (g)	xCO_2 (g)	$(\frac{y+1}{2})$ H_2O (l)
Initial volume/ cm^3	–	z	0	
After combustion and cooling/ cm^3	0	$z - 65$		
After reaction with NaOH/ cm^3	0	$z - 131 - 65$ (leftover O_2)	131 (formed)	

$$\text{Volume of } O_2 \text{ used} = z - (z - 131 - 65) = 196 \text{ cm}^3$$

$$\text{Amount of } O_2 \text{ used} = \frac{196 \times 10^{-3}}{22.7} = 8.63 \times 10^{-3} \text{ mol}$$

$$\text{Volume of } CO_2 \text{ formed} = 131 \text{ cm}^3$$

$$\text{Amount of } CO_2 \text{ formed} = \frac{131 \times 10^{-3}}{22.7} = 5.78 \times 10^{-3} \text{ mol}$$

Comparing ratio:

Mole ratio of $C_xH_yOH : CO_2$ is $1:x$.

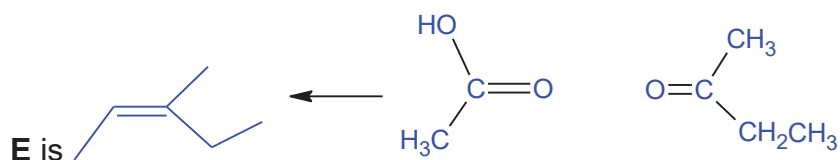
$$\text{Hence } \frac{1}{9.60 \times 10^{-4}} = \frac{x}{5.78 \times 10^{-3}} \therefore x = \frac{5.78 \times 10^{-3}}{9.60 \times 10^{-4}} = 6$$

$$\text{Comparing mole ratio of } C_xH_yOH \text{ and } O_2 = 1 : \frac{4x+y-1}{4}$$

$$\text{Since } x=6, \frac{1}{9.60 \times 10^{-4}} = \frac{\frac{4(6)+y-1}{4}}{8.63 \times 10^{-3}} \therefore y \approx 13$$

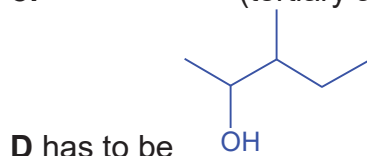
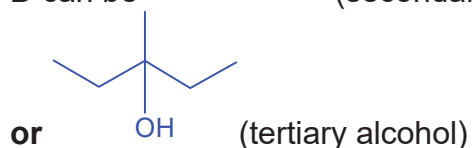
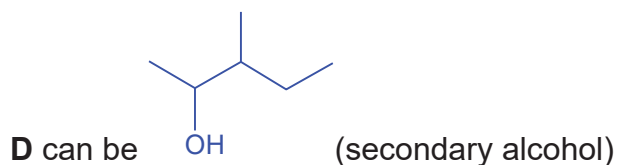
Hence **D** is $C_6H_{13}OH$.

(ii)



Alkene **E** is oxidised (•) by hot acidified conc. KMnO_4 to give ethanoic acid and butanone.

Note: Working backwards, **E** $\downarrow$ **D** (via addition of H-OH across the double bond)



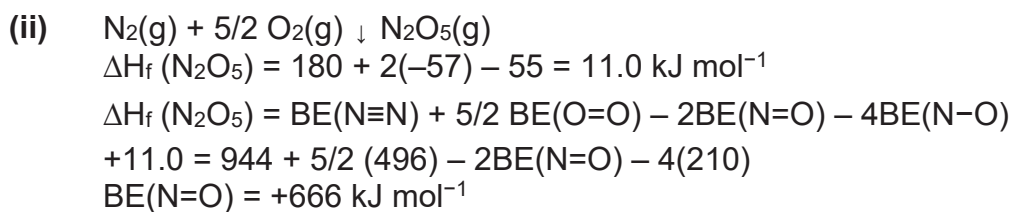
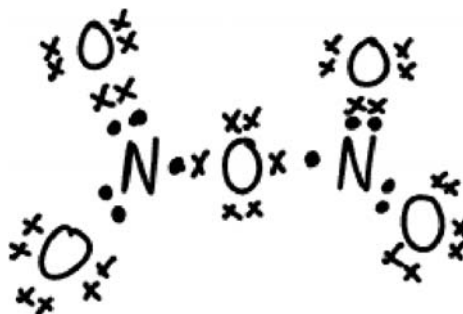
Since **D** can be oxidised (•) by $\text{K}_2\text{Cr}_2\text{O}_7$ it has to be a secondary alcohol.

Alkene **E** can be obtained by the elimination (•) of alcohol **D**.

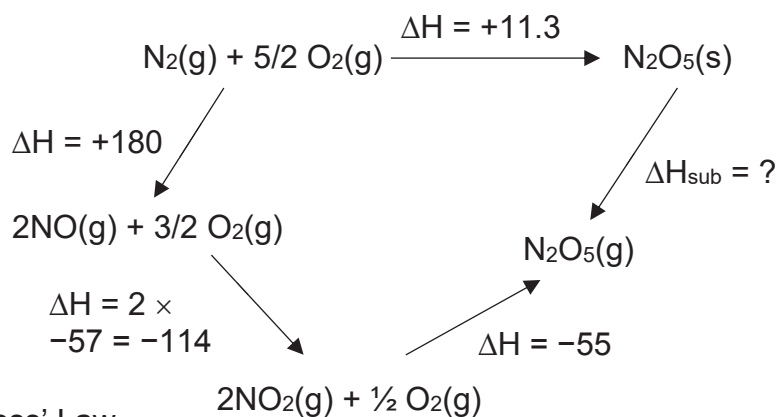
(iii) Yes. **E** can exhibit cis trans.

There is restricted rotation about the $\text{C}=\text{C}$ due to the presence of π bonding and there are 2 different groups attached to each of the carbon atom in the $\text{C}=\text{C}$.

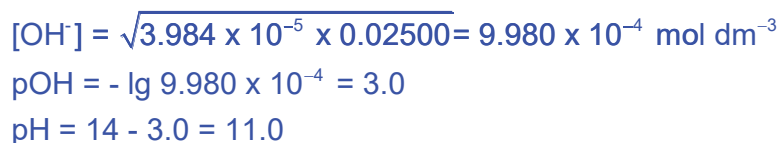
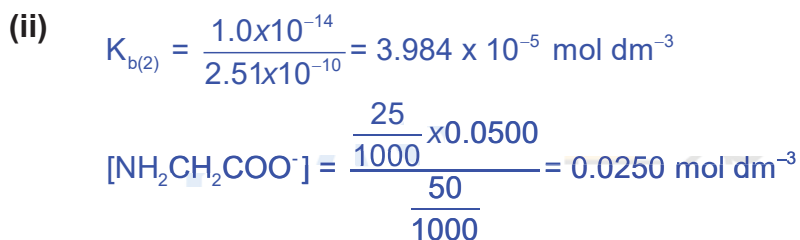
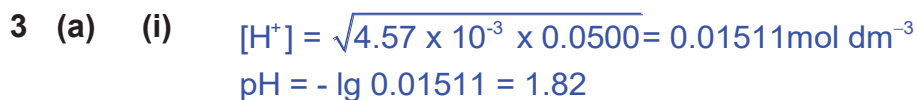
(e) (i)



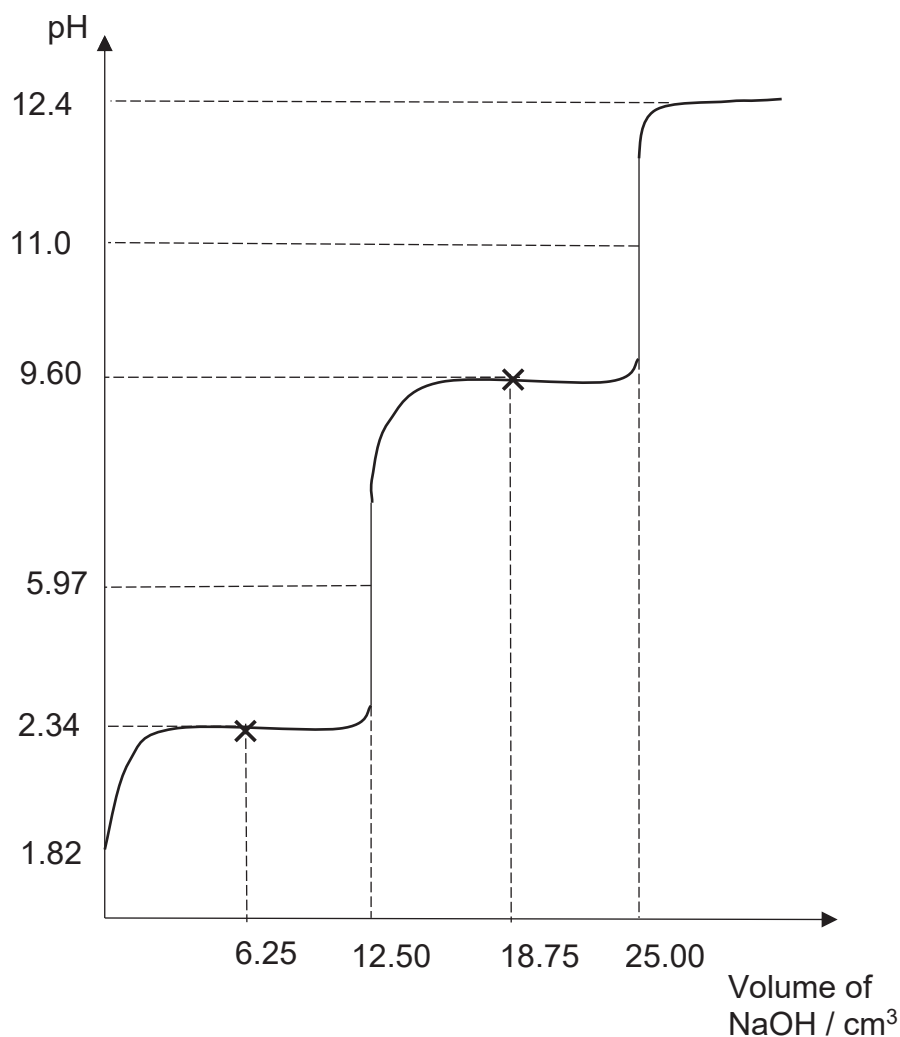
(iii)



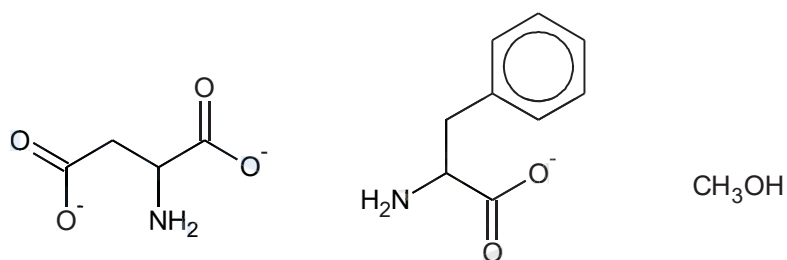
By Hess' Law,
 $\Delta H_{\text{sub}} = -11.3 + 180 + (-114) + (-55) = -0.300 \text{ kJ mol}^{-1}$



- (iii) pH at 1st M.B.C. = $pK_{a(1)} = 2.34$
 pH at 2nd M.B.C. = $pK_{a(2)} = 9.60$



(b)



- (c) (i) By Le Chatelier's principle, in the presence of high concentration of H^+ , the position of equilibrium shifts left to decrease the concentration of H^+ . Iron (II) sulfate dissolves, releasing free Fe^{2+} ions which are mobile and flushed through the intestine without being absorbed into the bloodstream.



$$K_f = \frac{[\text{Fe}(\text{CN})_6]^{4-}}{[\text{Fe}^{2+}][\text{CN}^-]^6}$$

Let the concentration of Fe^{2+} be $a \text{ mol dm}^{-3}$

$$10^{35} = \frac{0.65}{(a)(6a)^6}$$

$$a = 2.024 \times 10^{-6} \text{ mol dm}^{-3}$$

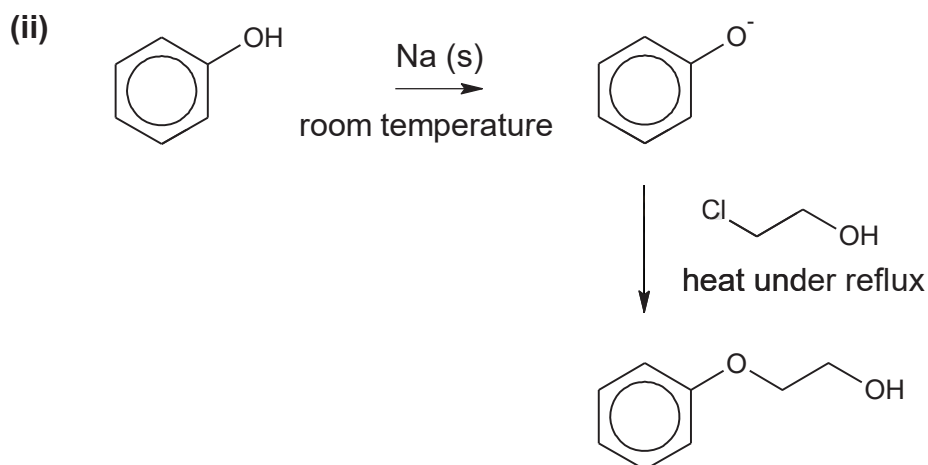
$$[\text{CN}^-] = 6 \times 2.024 \times 10^{-6} = 1.21 \times 10^{-5} \text{ mol dm}^{-3}$$

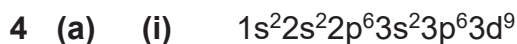
- (iii) K_f of complex formed between Fe^{2+} and CN^- is 10^{35} and CN^- binds strongly and reversibly with iron in haemoglobin. They are not easily released and prevent the haemoglobin from carrying oxygen. CN^- cannot be used as a ligand.

Since K_f of complex formed between Hg^{2+} and edta^{4-} is greater than that of complex formed between Hg^{2+} and cysteine ($10^{21} > 10^{14}$), edta^{4-} is able to displace Hg^{2+} from cysteine and forms a more stable complex.

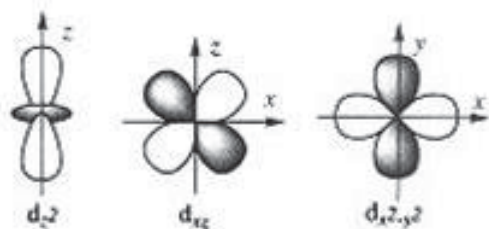
- (d) (i) 1,5-diaminopentane contain electron donating alkyl groups that increase the electron density on the N atom. This increases the availability of the lone pair of electrons on N atom for bonding with a proton. Hence, 1,5-diaminopentane is a stronger base than ammonia.

The p-orbital of N atom overlaps with the π -electron cloud of the benzene ring, causing the lone pair on N atom to be delocalised into the benzene ring. This decreases the availability of the lone pair of electrons on N atom for bonding with a proton. Hence, phenylamine is a weaker base than ammonia.





(ii)



(b) (i)

	E_{cell}/V
(1) $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$	+0.80
(2) $\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	+0.34
(3) $\text{Pb}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pb}$	-0.13

Reactions at anode:

(The operating voltage is carefully regulated so that it is sufficient for Cu to be oxidised.)

- Since $E_{\text{cell}}(\text{Cu}^{2+}/\text{Cu})$ is more positive than $E_{\text{cell}}(\text{Pb}^{2+}/\text{Pb})$, Pb will also be oxidised.
- Both Cu and Pb dissolve into the solution as cations and migrate to the cathode.
- Since $E_{\text{cell}}(\text{Ag}^+/\text{Ag})$ is more positive than $E_{\text{cell}}(\text{Cu}^{2+}/\text{Cu})$, Ag will not be oxidised.
- Ag drops off the electrode as the copper around dissolves, and fall to the bottom of the electrolytic tank to form anode sludge.

Reactions at cathode:

- Since $E_{\text{cell}}(\text{Cu}^{2+}/\text{Cu})$ is more positive than $E_{\text{cell}}(\text{Ni}^{2+}/\text{Ni})$ for (4), Cu^{2+} is preferentially reduced.
- Pb^{2+} remains in the solution.

(ii) $Q = It C$

$$I(\text{tx}60) = \frac{m_x - m_0}{63.5} (2) F [1]$$

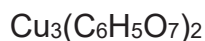
$$F = Le = \frac{63.5I(60t)}{2(m_x - m_0)}$$

$$L = \frac{63.5I(60t)}{2(m_x - m_0) \times 1.6 \times 10^{-19}}$$

$$= \frac{3810It}{3.2 \times 10^{-19}(m_x - m_0)} = \frac{1.19 \times 10^{22} It}{(m_x - m_0)} [1]$$

(c)

Element	Cu	C	H	O
Mass /g	33.55	25.30	1.75	39.40
Amount /mol	0.5283	2.108	1.75	2.462
Ratio	1	3.990	3.31	4.660
Simplest ratio	3	12	10	14



$$x = 3, y = 6, z = 5$$

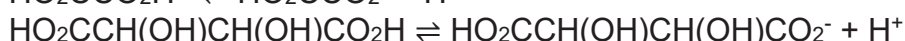
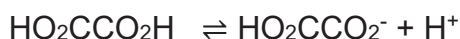
(d)

(i)

It can accept protons.

It can share/donate its lone pair of electrons on O^- .

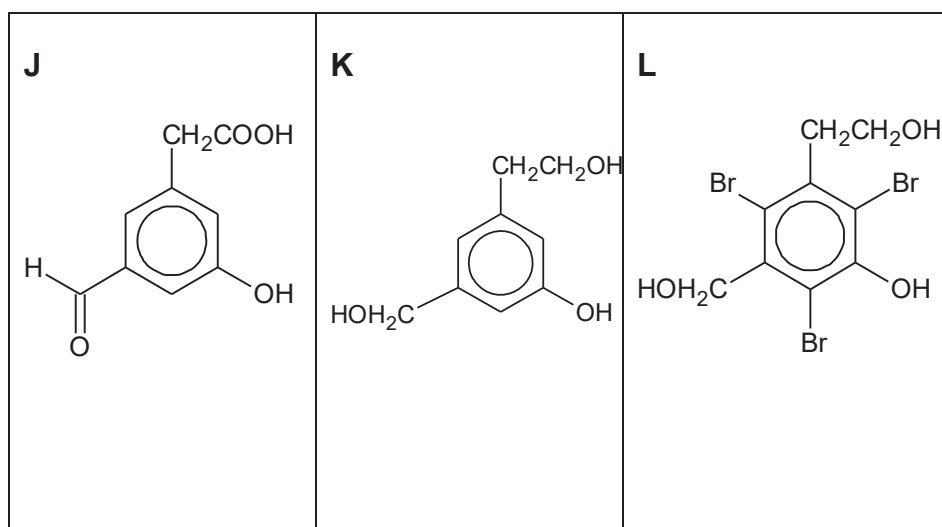
(ii)



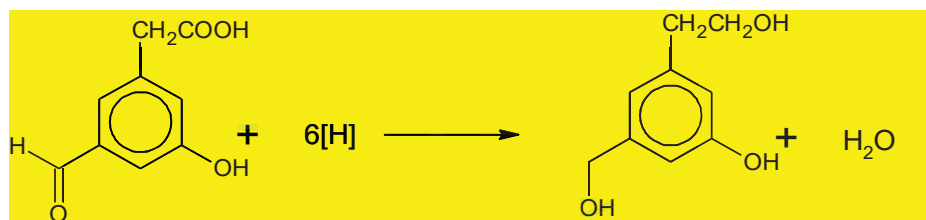
$\text{HO}_2\text{CCO}_2^-$ is more stable than $\text{HO}_2\text{CCH}(\text{OH})\text{CH}(\text{OH})\text{CO}_2^-$ as the negative charge on the O atom can be more effectively dispersed due to 2 electronegative O atoms are beside that carboxylate ion where there is only 1 O atom in the conjugate base of tartaric acid. Dissociation of ethanedioic is more favoured.

(e)

(i)



(ii)



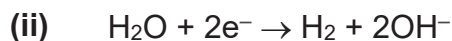
(iii)

2 observations: Orange bromine water decolourises, white ppt forms.

(iv)



- 5 (a) (i) The oxidation state of oxygen increases from -2 in CH_3COO^- to -1 in $\text{CH}_3\text{COO}\bullet$ hence the oxidation reaction took place at the anode.



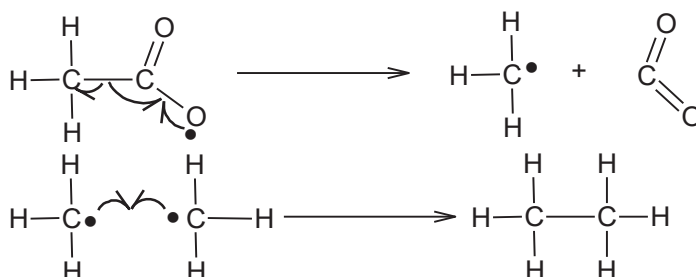
$$Q = It = 500 \times 10^{-3} \times 20 \times 60 = 600 \text{ C}$$

$$Q = nzF \Rightarrow 600 = n(2)(96500)$$

$$n(\text{H}_2) = 3.108 \times 10^{-3} \text{ mol}$$

$$v(\text{H}_2) = 3.108 \times 10^{-3} \times 24 = 0.0746 \text{ dm}^3 (= 74.6 \text{ cm}^3)$$

(iii)



(b) (i)

$$PV = \frac{m}{M}RT$$

$$M = \frac{mRT}{PV}$$

$$= \frac{(0.30)(8.31)(300)}{(101325) \times (168 \times 10^{-6})}$$

$$= 43.9 \text{ g mol}^{-1} \Rightarrow M_r = 43.9$$

Let molecular formula of X be $\text{C}_n\text{H}_{2n+2}$ do include this line to introduce what's n.

$$12(n) + 2n + 2 = 44$$

$$n = 3 \Rightarrow \text{X is C}_3\text{H}_8$$

- (ii) Intermolecular forces between the gas particles is negligible/insignificant.

The volume of gas particles is insignificant/negligible compared to volume of container.

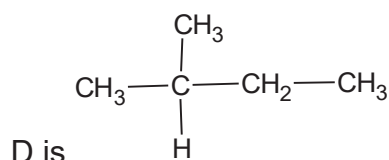
- (iii) X is C_3H_8 hence its structural formula is $\text{CH}_3\text{CH}_2\text{CH}_3$. Since it is unsymmetrical R-R', it is formed by two different alkyl radicals ($\text{CH}_3\bullet$ from ethanoate and another $\text{R}\bullet$ from salt of acid A).

Hence, salt of acid W produced $\text{CH}_3\text{CH}_2\bullet$ in step III.

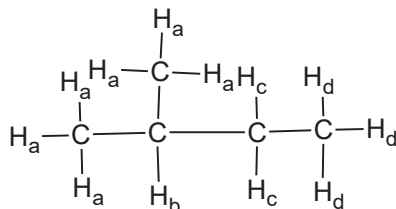
Alkane Y (R'-R') is $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

Acid W contains an additional COOH (lost during step II decarboxylation) attached to $\text{CH}_3\text{CH}_2\bullet$ hence it is $\text{CH}_3\text{CH}_2\text{COOH}$

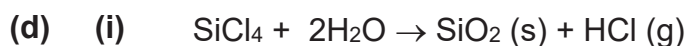
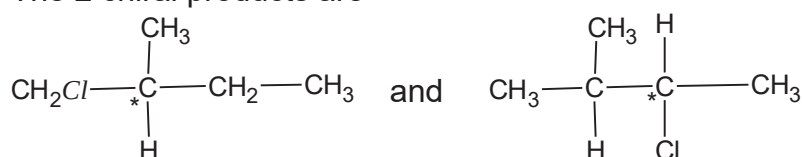
(c)



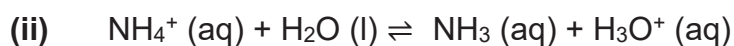
There are four different types of H in different chemical environment in D as shown below.



The 2 chiral products are



SiCl_4 undergoes hydrolysis to form a strongly acidic solution of pH 1-2. The solution turns red.



Ammonium chloride is an acidic salt. The conjugate acid NH_4^+ hydrolyses in water to form a weakly acidic solution of pH 5 (approx. mid-way of pH change at eq pt for WB-SA titration pH3-7). The solution turns purple/pink.